

printpdf - Print to PDF Tool

A versatile tool for converting various content sources to PDF format using multiple PDF generation engines.

Features

Input Sources

- **Local Markdown files:** Direct file paths
- **HTTP/HTTPS URLs:** Remote Markdown files
- **GitHub files:** Direct links to files in GitHub repositories (including private repos with authentication)
 - Blob URLs: `https://github.com/owner/repo/blob/branch/file.md`
 - Raw URLs: `https://raw.githubusercontent.com/owner/repo/branch/file.md`
 - Commit URLs: `https://github.com/owner/repo/blob/commit-sha/file.md`
- **Web pages:** Regular web pages (processed with Mozilla Readability for clean content)

PDF Converters

The tool supports multiple PDF conversion engines and will try all of them by default:

- **Typst:** Modern typesetting system (automatically downloaded if not present)
- **Prince XML:** Commercial HTML-to-PDF converter (requires manual installation)
- **WeasyPrint:** Python-based HTML-to-PDF converter (requires `pip install weasyprint`)

Each converter has different strengths:

- **Typst:** Best for documents with modern typography
- **Prince:** Professional output with excellent CSS support
- **WeasyPrint:** Open-source, good for simple documents

Installation

Build from source

```
cd printpdf
go build -o printpdf ./cmd
```

Using mise

```
cd printpdf
mise run build
```

Usage

Basic usage

```
# Convert a local Markdown file
printpdf document.md

# Convert from a URL
printpdf https://example.com/document.md

# Convert a GitHub file (public repository)
printpdf https://github.com/owner/repo/blob/main/README.md

# Convert a GitHub file (private repository - requires GITHUB_TOKEN)
export GITHUB_TOKEN=your_token_here
printpdf https://github.com/owner/repo/blob/main/README.md

# Convert a web page (uses Readability)
printpdf https://example.com/article
```

Options

```
# Specify output directory
printpdf -o ./output document.md

# Use specific converters (comma-separated)
printpdf -converters typst,weasyprint document.md
```

```
# Use only one converter
printpdf -converters typst document.md
```

GitHub Authentication

For private repositories, set your GitHub token:

```
export GITHUB_TOKEN=ghp_your_token_here
```

Or use the GitHub CLI:

```
gh auth login
```

The tool will automatically use your GitHub token if available in the environment.

Output

The tool generates PDF files in the specified output directory (current directory by default):

- `output-typst.pdf` - Generated with Typst
- `output-prince.pdf` - Generated with Prince XML
- `output-weasyprint.pdf` - Generated with WeasyPrint

This allows you to compare the output quality of different converters and choose the best one for your needs.

Automatic Tool Download

The tool automatically manages PDF converter downloads:

- **Typst**: Downloaded automatically from GitHub releases
- **Prince**: Must be installed manually (commercial software)
- **WeasyPrint**: Must be installed via pip (Python required)

Downloaded tools are cached in `~/.cache/printpdf/` for future use.

Examples

See the `samples/` directory for example PDF outputs generated from various sources.

Requirements

- Go 1.24.7 or later (for building)
- Internet connection (for downloading converters and fetching remote content)

Optional:

- Python 3 with pip (for WeasyPrint)
- Prince XML license (for Prince converter)

Development

Run tests

```
mise run test
```

Format code

```
mise run fmt
```

Build

```
mise run build
```

Architecture

The tool is organized into several packages:

- `cmd/` - Main entry point
- `pkg/fetcher/` - Content fetching from various sources
- `pkg/converter/` - PDF conversion engines

- `pkg/downloader/` - Tool download and caching

License

See [LICENSE](#) in the repository root.